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SEQUENTIAL CIRCUIT BASED OSCILLATOR

Abstract

An apparatus includes a first edge-triggered storage device that has a first clock input, a first data input, and a first data output. The apparatus also includes a first feedback path from the first data output to the first clock input. The apparatus further includes a second edge-triggered storage device that has a second clock input, a second data input, and a second data output. The second clock input is coupled to the first data output. The apparatus additionally includes a second feedback path from the second data output to the second clock input, as well as a third feedback path from the second data output to the first clock input.

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Background/Summary

TECHNICAL FIELD

[0001] The technology discussed below relates generally to oscillation circuits and, more

particularly, to a ring oscillator implemented using sequential circuits.

INTRODUCTION

[0002] Electronic device technologies have seen explosive growth over the past several years. For example, growth in wireless communication technologies and network technologies has been fueled by more advanced hardware and software. Service providers are now able to offer their customers an ever-expanding array of features and services, and provide users with unprecedented levels of access to information, resources, and communication services. To keep pace with these service enhancements, electronic devices (e.g., cellular phones, computers, servers, tablets, laptops, etc.) have higher performance (e.g., higher processing rates, higher transmission rates, higher computational rates, etc.) and complexity than ever.

[0003] With this increase in performance and complexity, the power consumption of electronic devices has generally increased. In addition, it is desirable to obtain the maximum performance from an electronics device as efficiently as possible. Therefore, there is an ongoing need for monitoring the silicon processing condition for performance and power-related improvements in electronic devices. Ring oscillators are commonly used as process monitors in state-of-the-art chip designs, which can provide real-time information to control the power supply of high performance low power circuits.

BRIEF SUMMARY OF SOME EXAMPLES

[0004] The following presents a summary of one or more aspects of the present disclosure, in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated features of the disclosure and is intended neither to identify key or critical elements of all aspects of the disclosure nor to delineate the scope of any or all aspects of the disclosure. Its sole purpose is to present some concepts of one or more aspects of the disclosure in a form as a prelude to the more detailed description that is presented later.

[0005] In various aspects of the disclosure, an apparatus includes a first edge-triggered storage device that has a first clock input, a first data input, and a first data output. The apparatus also includes a first feedback path from the first data output to the first clock input. The apparatus further includes a second edge-triggered storage device that has a second clock input, a second data input, and a second data output. The second clock input is coupled to the first data output. The apparatus additionally includes a second feedback path from the second data output to the second clock input, as well as a third feedback path from the second data output to the first clock input.

[0006] In various aspects of the disclosure, a ring oscillator includes a first edge-triggered storage device that has a first clock input, a first data input, and a first data output. The ring oscillator also includes a first logic circuit that has a first input, a second input, and a first output. The first input is coupled to the first data output of the first edge-triggered storage device. The second input is coupled to a reset circuit. The first output is coupled to the first data input of the first edge-triggered storage device. The ring oscillator further includes a first multiplexer that has a first multiplexer input, a second multiplexer input, a control input, and a multiplexer output. The first multiplexer input is coupled to the first data output of the first edge-triggered storage device. The control input is coupled to the reset circuit. The ring oscillator additionally includes a second logic circuit that has a third input, a fourth input, and a second output. The third input is coupled to the multiplexer output. The second output is coupled to the first clock input of the first edge-triggered storage device.

[0007] These and other aspects of the disclosure will become more fully understood upon a review of the detailed description which follows. Other aspects, features, and examples of the present disclosure will become apparent to those of ordinary skill in the art, upon reviewing the following description of specific, example aspects of the present disclosure in conjunction with the accompanying figures. While features of the present disclosure may be discussed relative to certain examples and figures below, all examples of the present disclosure can include one or more of the advantageous features discussed herein. In other words, while one or more examples may be

discussed as having certain advantageous features, one or more of such features may also be used in accordance with the various examples of the disclosure discussed herein. In similar fashion, while example aspects may be discussed below as device, system, or method examples it should be understood that such example aspects can be implemented in various devices, systems, and methods.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 illustrates an example of a system-on-a-chip (SOC) that may be adapted in accordance with certain aspects of the disclosure.

[0009] FIG. 2 illustrates an example of an apparatus that includes a voltage controller for scaling power supply voltages in accordance with certain aspects of the disclosure.

[0010] FIG. 3 illustrates an example of a ring oscillator (RO) that includes sequential circuitry such as edge-triggered memory devices in accordance with certain aspects of the disclosure.

[0011] FIG. 4 illustrates another example of a RO in accordance with certain aspects of the disclosure.

[0012] FIG. 5 illustrates an example of an initial state of the RO of FIG. 4.

[0013] FIG. 6 illustrates an example of a first operational state of the RO of FIG. 4.

[0014] FIG. 7 illustrates an example of a second operational state of the RO of FIG. 4.

[0015] FIG. 8 illustrates an example of a third operational state of the RO of FIG. 4.

[0016] FIG. 9 illustrates an example of a fourth operational state of the RO of FIG. 4.

[0017] FIG. 10 illustrates an example of a fifth operational state of the RO of FIG. 4.

[0018] FIG. 11 illustrates an example of a sixth operational state of the RO of FIG. 4.

[0019] FIG. 12 illustrates an example timing diagram for the RO of FIG. 4.

[0020] FIG. 13 illustrates multi-stage RO in accordance with certain aspects of the disclosure.

[0021] FIG. 14 illustrates an example ring oscillator method in accordance with certain aspects of the disclosure.

DETAILED DESCRIPTION

[0022] The detailed description set forth below in connection with the appended drawings is intended as a description of various configurations and is not intended to represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

[0023] While aspects and examples are described in this application by illustration to some examples, those skilled in the art will understand that additional implementations and use cases may come about in many different arrangements and scenarios. Innovations described herein may be implemented across many differing platform types, devices, systems, shapes, sizes, and packaging arrangements. For example, aspects and/or uses may come about via integrated chip examples and other non-module-component based devices (e.g., end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, artificial intelligence-enabled (AI-enabled) devices, etc.). While some examples may or may not be specifically directed to use cases or applications, a wide assortment of applicability of described innovations may occur. Implementations may range a spectrum from chip-level or modular components to non-modular, non-chip-level implementations and further to aggregate, distributed, or original equipment manufacturer (OEM) devices or systems incorporating one or more aspects of the described innovations. In some practical settings, devices incorporating

described aspects and features may also necessarily include additional components and features for implementation and practice of claimed and described examples. For example, transmission and reception of wireless signals necessarily includes a number of components for analog and digital purposes (e.g., hardware components including antenna, radio frequency (RF) chains, power amplifiers, modulators, buffer, processor(s), interleaver, adders/summers, etc.). It is intended that innovations described herein may be practiced in a wide variety of devices, chip-level components, systems, distributed arrangements, disaggregated arrangements (e.g., base station and/or UE), end-user devices, etc., of varying sizes, shapes, and constitution.

[0024] The terms “computing device” and “mobile device” are used interchangeably herein to refer to any one or all of servers, personal computers, smartphones, cellular telephones, tablet computers, laptop computers, notebooks, ultrabooks, palm-top computers, personal data assistants (PDAs), wireless electronic mail receivers, multimedia Internet-enabled cellular telephones, Global Positioning System (GPS) receivers, wireless gaming controllers, and similar personal electronic devices which include a programmable processor. While the various aspects are particularly useful in mobile devices (e.g., smartphones, laptop computers, etc.), which have limited resources (e.g., processing power, battery, size, etc.), the aspects are generally useful in any computing device that may benefit from improved processor performance and reduced energy consumption.

[0025] The term “multicore processor” is used herein to refer to a single integrated circuit (IC) chip or chip package that contains two or more independent processing units or cores (e.g., central processing unit (CPU) cores, etc.) configured to read and execute program instructions. The term “multiprocessor” is used herein to refer to a system or device that includes two or more processing units configured to read and execute program instructions.

[0026] The term “system on chip” (SoC) is used herein to refer to a single integrated circuit (IC) chip that contains multiple resources and/or processors integrated on a single substrate. A single SoC may contain circuitry for digital, analog, mixed-signal, and radio-frequency functions. A single SoC may also include any number of general purpose and/or specialized processors (digital signal processors (DSPs), modem processors, video processors, etc.), memory blocks (e.g., read only memory (ROM), random access memory (RAM), flash, etc.), and resources (e.g., timers, voltage regulators, oscillators, etc.), any or all of which may be included in one or more cores.

[0027] Process technology employed to manufacture semiconductor devices, including IC devices is continually improving. Process technology includes the manufacturing methods used to make IC devices and defines transistor size, operating voltages and switching speeds. Features that are constituent elements of circuits in an IC device may be referred as technology nodes and/or process nodes. The terms technology node, process node, process technology may be used to characterize a specific semiconductor manufacturing process and corresponding design rules. Faster and more power-efficient technology nodes are being continuously developed through the use of smaller feature size to produce smaller transistors that enable the manufacture of higher-density ICs.

[0028] ICs typically provide multiple voltage domains for power saving purposes. For example, higher voltage domains provide power at higher voltage levels than lower voltage domains. Higher voltage domains are sometimes needed for interfacing with external devices, while core logic circuits can generally operate at the lower voltage levels available in lower voltage domains. For the purposes of this disclosure, a thick-oxide transistor may refer to a transistor that has a gate oxide thickness sufficient to enable the transistor to withstand and operate at the higher voltage levels in higher voltage domains and a thin-oxide transistor may refer to a transistor that has a gate oxide thickness that is insufficient to avoid electrical overstress when the transistor spans a higher voltage level in the higher voltage domains. In certain examples disclosed herein, a thin-oxide transistor may be rated for voltages up to 0.6 volts and a thick-oxide transistor may be rated for voltages greater than 0.6 volts and thick-oxide transistors may be used in a higher a high-voltage domain that provides power at 1.2 volts.

[0029] Certain circuits are illustrated in this disclosure as being implemented using combinations

of P-type metal-oxide-semiconductor (PMOS) transistors and N-type metal-oxide-semiconductor (NMOS) transistors. These circuits are provided by way of example only, and it is contemplated that the concepts disclosed herein can be implemented in circuits that use different combinations of NMOS and PMOS transistors. Circuits that include NMOS and PMOS transistors are typically coupled to the rails of a power supply. The power supply provides a current that flows from a higher voltage rail to a lower voltage rail. A rail may include some combination of conductors, wires, connectors and other types of interconnect. For the purposes of this description, the higher voltage rail may be referenced as “VDD” or “VDD” and the lower voltage rail may be referred to as Ground. In some implementations, power may be provided to certain circuits through more than two rails.

[0030] For purposes of illustration, FIG. 1 illustrates examples of components and interconnections in a system-on-chip (SoC) **100**, including a memory interface/bus **126**, that may be suitable for implementing certain aspects of the present disclosure. The SoC **100** may include a number of heterogeneous processors, such as a central processing unit (CPU) **102**, a modem processor **104**, a graphics processor **106**, and an application processor **108**. Each processor **102**, **104**, **106**, **108**, may include one or more cores, and each processor/core may perform operations independent of the other processors/cores. The processors **102**, **104**, **106**, **108** may be organized in close proximity to one another (e.g., on a single substrate, die, integrated chip, etc.) so that the processors may operate at a much higher frequency/clock rate than would be possible if the signals were to travel off-chip. The proximity of the cores may also allow for the sharing of on-chip memory and resources (e.g., voltage rails), as well as for more coordinated cooperation between cores.

[0031] The SoC **100** may include interface circuits that support inter-processor communication and/or communication between one or more of the processors **102**, **104**, **106**, **108**, memory, wireless modems, and other peripherals. Many of these interface circuits are based on serializer/deserializer (SerDes) circuits that enable transmission and reception of data over a communication link. SerDes circuits may be used to support high-speed and low-speed communication.

[0032] The SoC **100** may include system components and resources **110** for managing sensor data, analog-to-digital conversions, and/or wireless data transmissions, and for performing other specialized operations (e.g., decoding high-definition video, video processing, etc.). System components and resources **110** may also include components such as voltage regulators, voltage controllers, sensors, oscillators, phase-locked loops (PLLs), peripheral bridges, data controllers, system controllers, access ports, timers, registers, and/or other similar components used to support the processors and software clients running on the computing device. The system components and resources **110** may also include circuitry for interfacing with peripheral devices, such as cameras, electronic displays, wireless communication devices, external memory chips, etc.

[0033] The SoC **100** may further include a Universal Serial Bus (USB) or other serial bus controller **112**, one or more memory controllers **114**, and a centralized resource manager (CRM) **116**. The SoC **100** may also include an input/output module (not illustrated) for communicating with resources external to the SoC, each of which may be shared by two or more of the internal SoC components.

[0034] The processors **102**, **104**, **106**, **108** may be interconnected to the USB controller **112**, the memory controller **114**, system components and resources **110**, CRM **116**, and/or other system components via an interconnection/bus module **122**, which may include an array of reconfigurable logic gates and/or implement a bus architecture. Communications may also be provided by advanced interconnects, such as high-performance networks on chip.

[0035] The interconnection/bus module **122** may include or provide a bus mastering system configured to grant SoC components (e.g., processors, peripherals, etc.) exclusive control of the bus (e.g., to transfer data in burst mode, block transfer mode, etc.) for a set duration, number of operations, number of bytes, etc. In some cases, the interconnection/bus module **122** may

implement an arbitration scheme to prevent multiple master components from attempting to drive the bus simultaneously.

[0036] The memory controller **114** may be a specialized hardware module configured to manage the flow of data to and from a memory **124** via the memory interface/bus **126**. In some examples, the memory controller **114** includes one or more processors configured to perform read and write operations with the memory **124**. Examples of processors include microprocessors, microcontrollers, digital signal processors (DSPs), field programmable gate arrays (FPGAs), programmable logic devices (PLDs), state machines, gated logic, discrete hardware circuits, and other suitable hardware configured to perform the various functionality described throughout this disclosure.

[0037] Memory technologies described herein may be suitable for storing instructions, programs, control signals, and/or data for use in or by a computer or other digital electronic device. Any references to terminology and/or technical details related to an individual type of memory, interface, standard, or memory technology are for illustrative purposes only, and not intended to limit the scope of the claims to a particular memory system or technology unless specifically recited in the claim language. Mobile computing device architectures have grown in complexity, and now commonly include multiple processor cores, SoCs, co-processors, functional modules including dedicated processors (e.g., communication modem chips, GPS receivers, etc.), complex memory systems, intricate electrical interconnections (e.g., buses and/or fabrics), and numerous other resources that execute complex and power intensive software applications (e.g., video streaming applications, etc.).

[0038] In general, it is desirable that an electronic device (e.g., the SoC **100** of FIG. **1** or some other type of electronic device) provide the highest possible performance while keeping power consumption as low as possible. For example, excessive power consumption may lead to shorter battery life and/or heat-related issues (e.g., which can adversely affect performance and/or cost).

[0039] In some examples, an electronic device may be capable of adjusting a power supply voltage level depending on operating conditions and/or other factors. As one example, an electronic device may transmit lower frequency signals at higher voltage levels than higher frequency signals. To support such power optimization, an electronic device may include circuitry to scale the power supply voltage (e.g., VDD) provided to various circuits in the electronic device.

[0040] As another example, process variations within an integrated circuit die may cause some circuits within the integrated circuit to operate with different delays at a given power supply voltage. In this case, it may be desirable to scale the power supply voltage according to the delay characteristics of the different circuits.

[0041] The SoC **100** may include ring oscillator (RO) sensors **128** that provide feedback for VDD scaling. These and other aspects of VDD scaling will be discussed with reference to FIG. **2**.

[0042] FIG. **2** illustrates an example of an apparatus **202** that includes a voltage controller **204** configured to scale power supply voltages provided to different circuits within apparatus **202**. The apparatus **202** may be an integrated circuit (IC) in some examples or a device that includes at least one IC in other examples. Examples of such a device include a computer, a server, a user equipment, a base station, a mobile device, a cellular (cell) phone, a smart phone, a session initiation protocol (SIP) phone, a laptop, a personal computer (PC), a notebook, a netbook, a smartbook, a tablet, a personal digital assistant (PDA), and a broad array of embedded systems, e.g., corresponding to an Internet of Things (IoT). A mobile apparatus may be, for example, an automotive or other transportation vehicle, a remote sensor or actuator, a robot or robotics device, a satellite radio, a global positioning system (GPS) device, an object tracking device, a remote control device, a consumer and/or wearable device, such as eyewear, a wearable camera, a virtual reality device, a smart watch, a health or fitness tracker, a digital audio player (e.g., MP3 player), a camera, a game console, etc. A mobile apparatus may be a digital home or smart home device such as a home audio, video, and/or multimedia device, an appliance, a vending machine, intelligent

lighting, a home security system, a smart meter, etc. A mobile apparatus may be a smart energy device, a security device, a solar panel or solar array, a municipal infrastructure device controlling electric power (e.g., a smart grid), lighting, water, etc., an industrial automation and enterprise device, a logistics controller, agricultural equipment, etc. Still further, a mobile apparatus may provide for connected medicine or telemedicine support, i.e., health care at a distance. Telehealth devices may include telehealth monitoring devices and telehealth administration devices, the communicated information of which may be given preferential treatment or prioritized access over other types of information, e.g., in terms of prioritized access for transport of critical service data, and/or relevant quality of service (QOS) for transport of critical service data. A base station may be, for example, a network element in a radio access network responsible for radio transmission and reception in one or more service sets. In different technologies, standards, or contexts, an access point may variously be referred to by those skilled in the art as a base station, a base transceiver station (BTS), a radio base station, a radio transceiver, a transceiver function, a basic service set (BSS), an extended service set (ESS), a Node B (NB), an eNode B (eNB), a gNode B (gNB), a transmission and reception point (TRP), or some other suitable terminology.

[0043] Different sections of the apparatus **202** include different logic circuits (logic). For example, a first cell **206** of an IC may include first logic **208** and a second cell **210** of the IC may include second logic **212**. As another example, a first core **214** of the apparatus **202** may include third logic **216** and a second core **218** of the apparatus **202** may include fourth logic **220**. Thus, in general, the apparatus **202** may include various logic (e.g., fifth logic **222**) at various locations within the apparatus **202**.

[0044] In some examples, the voltage controller **204** scales power supply voltages based on IC process variations at different sections of the apparatus **202** (e.g., process variations across an IC die). The voltage controller **204** may scale power supply voltages based on other factors in other examples.

[0045] IC process variations may result in different operating characteristics for circuits in different section of the IC die. For example, circuits in one section may have different delay characteristics than circuits in another section when the circuits are powered by the same power supply voltage.

[0046] The apparatus **202** may include sensor circuits such as ring oscillators to determine (e.g., estimate) relative process variations. For example, ring oscillators implemented at different sections of the apparatus **202** subject to process variations may oscillate at different frequencies (e.g., due to different delay characteristics of the components of the ring oscillators in the different sections).

[0047] In the example, of FIG. 2, the apparatus **202** includes a first ring oscillator (RO) **224**, a second RO **226**, a third RO **228**, a fourth RO **230**, and a fifth RO **232**. A different number of ROs may be used in other examples.

[0048] Each of the ROs in FIG. 2 is implemented near logic subject to power supply voltage scaling. The first RO **224** is implemented near the first logic **208**, the second RO **226** is implemented near the second logic **212**, the third RO **228** is implemented near the third logic **216**, the fourth RO **230** is implemented near the fourth logic **220**, and the fifth RO **232** is implemented near the fifth logic **222**.

[0049] A given RO may be expected to have similar process characteristics as the corresponding nearby logic. Thus, any difference in process variations between the first logic **208** and the second logic **212** may be comparable to (e.g., the same as or almost the same as) difference in process variations between the first RO **224** and the second RO **226**.

[0050] As discussed above, a difference between the frequency of the first RO **224** and the frequency of the second RO **226** may be indicative of a difference in process variations between the first RO **224** and the frequency of the second RO **226**. Consequently, this difference in frequency may be indicative of a difference in process variations between the first logic **208** and the second logic **212**.

[0051] The voltage controller **204** may therefore use feedback of this frequency information to scale the power supply voltages for the first logic **208**, the second logic **212**, the third logic **216**, the fourth logic **220**, and the fifth logic **222**. As shown in FIG. 2, the voltage controller **204** receives a first signal **234** from the first RO **224**, a second signal **236** from the second RO **226**, a third signal **238** from the third RO **228**, a fourth signal **240** from the fourth RO **230**, and a fifth signal **242** from the fifth RO **232**. A comparator **244** (or some other circuitry) generates a signal or signals corresponding to the difference or differences between two or more of the received signals. For example, the comparator **244** may include a phase-locked loop (PLL) or delay-locked loop (DLL) circuit that generates an error signal indicative of a difference between a nominal frequency and the frequency of a received signal.

[0052] In the example of FIG. 2, a voltage scaler **246** scales one or more power supply voltages based on the signal or signal from the comparator **244**. For example, the voltage scaler **246** may apply a different scaling factor to a first VDD rail **248** for the first cell **206** than it applies to a second VDD rail **250** for the second cell **210** and the second core **218**. Here, the difference between the scaling factors may be based on differences between the frequencies of the first signal **234**, the second signal **236**, and the fourth signal **240**. As another example, the voltage scaler **246** may apply a different scaling factor to a third VDD rail **252** for the first core **214** and the fifth logic **222** than it applies to the second VDD rail **250**. Here, the difference between the scaling factors may be based on differences between the frequencies of the third signal **238**, the fourth signal **240**, and the fifth signal **242**. Other techniques for scaling power supply voltages may be used in other examples.

[0053] Conventionally, ROs used for power supply voltage scaling above are implemented using combinational logic. For example, ROs may be constructed from logic devices such as inverters, buffer, NAND gates, and NOR gates. Thus, these ROs may effectively characterize the effects of process variations on such circuits.

[0054] However, many circuits include both combinational logic and sequential logic (e.g., flip flops, registers, etc.). For example, 40% of a CPU or a graphics processing unit (GPU) may be sequential logic in some cases.

[0055] The disclosure relates in some aspects to an RO that incorporates sequential logic. For example, an apparatus may use a sequential RO circuit topology to represent a sequential cell for VDD scaling feedback. Such an RO may be referred to herein as a sequential RO, for convenience. A feedback scheme that uses a sequential ROs may better characterize the effects of process variations within the apparatus, as compared to feedback schemes that use ROs that only employ combinatorial logic, since the sequential ROs may better reflect the process variations associated the sequential logic in the apparatus. In addition, the use of such sequential ROs may be used to correlate a design model with the actual IC generated based on the design model.

[0056] Referring to FIG. 3, the disclosure relates in some aspects to an RO **300** that includes sequential circuitry such as edge-triggered memory devices. As used herein, an edge-triggered memory device refers to a memory device that is activated (e.g., latches the value at an input) in response to a rising edge or a falling edge at a clock input of the memory device. For example, for a rising-edge edge-triggered memory device, if a value of 1 is present at the data input of the memory device, the memory device will latch that value when the signal at the clock input of the memory device transitions from a low state to a high state (i.e., a rising edge of a clock signal). Examples of edge-triggered memory devices include flip flops (FFs), registers, and so on.

[0057] In the example of FIG. 3, the edge-triggered memory devices are so-called D FFs. D FFs have a data input D (or d), a data output Q (or q), and a clock input CLK (or clk). The D FFs of FIG. 3 could be replace with other types of edge-triggered memory devices in other examples.

[0058] For purposes of illustration, the RO **300** is shown as including three stages. The first stage includes a first flip flop (FF) **302** (e.g., a D flip flop (DFF)), the second stage includes a second FF **304**, and the third stage includes a third FF **306**. A different number of stages may be used in other examples. As one example, an RO may include 20 or more stages in some implementations.

[0059] An overview of the operation of a given stage of the RO **300** follows. Oscillation of the RO **300** is controlled by an enable signal (ro_enable). The ro_enable signal transitioning to a high state (e.g., a logic “1”) triggers the clock input of the first stage of the RO **300**, thereby causing the first stage to propagate its D input (initially a high state) to its Q output. This transition in the Q output of the first stage to a high state causes the following action. First, this transition triggers a rising clock (0->1) for the next stage whereby the next stage propagate its D input (initially a high state) to its Q output. In addition, the transition of the Q output of the first stage to a high state causes the clock for the first stage to be reset back to zero (1->0) to set up the opposite data (master latch) required for the next phase of oscillation. Also, the transition of the Q output of the first stage to a high state causes the opposite value (low state) to be presented to the D input of the first stage as required for the next phase of oscillation. These actions are propagated through each stage of the RO **300**. When the corresponding signal propagates to the final stage of the RO **300**, this signal is fed back to the first stage of the RO **300** to trigger another clock rising edge (0->1) for first stage, thereby starting the next phase of oscillation.

[0060] From the above, it may be observed that the RO **300** is a self-timed RO. For example, after a clock rising edge triggers a flip flop, the clock signal is automatically reset back to a low state (e.g., a logic “0”) to be ready for the next clock rising edge. This is accomplished using the output of a downstream node in the RO **300** that provides feedback to reset the clock back to the low state.

[0061] The components and operation of the RO **300** will now be discussed in more detail.

[0062] At each stage of the RO **300**, an inverter (and, optionally, a buffer) feeds back the Q output of the FF for that stage to the D input of the FF. This causes the Q output of the FF to oscillate between the high state and the low state with each incoming clock pulse at the clock input of the FF. Thus, at the first FF **302**, the Q output at a signal path **308** is fed back to the D input via a first buffer **310** and a first inverter **312**. Similarly, at the second FF **304**, the Q output at a signal path **314** is fed back to the D input via a second buffer **316** and a second inverter **318**. Also, at the third FF **306**, the Q output at a signal path **320** is fed back to the D input via a third buffer **322** and a third inverter **324**.

[0063] Also at each stage of the RO, an exclusive OR (XOR) gate feeds back the Q output of the FF to the CLK input of the FF. As discussed in more detail below, this automatically resets the CLK output back to a low state. This feature enables the FF to be repeatedly clocked by a succession of rising edges at the CLK input whereby, after each rising edge, the CLK input is automatically reset back to the low state. At the first FF **302**, the Q output at the signal path **308** is fed back through a first XOR **326** to the CLK input via a signal path **328**. Similarly, at the second FF **304**, the Q output at the signal path **314** is fed back through a second XOR **330** to the CLK input via a signal path **332**. Also, at the third FF **306**, the Q output at the signal path **320** is fed back through a third XOR **334** to the CLK input via a signal path **336**.

[0064] An ro_enable signal enables oscillation of the RO **300**. Initially, it is assumed that the ro_enable signal and the Q outputs of the first FF **302**, the second FF **304**, and the third FF **306** are at the low state (e.g., as discussed in more detail below). Thus, the D inputs of the first FF **302**, the second FF **304**, and the third FF **306** are at the high state. In this case, the output of a fourth XOR **340** on a signal path **342** will also be at the low state since both inputs of the fourth XOR **340** are at the low state at this point. Similarly, the output (the signal path **328**) of the first XOR **326** will be at the low state since both inputs of the first XOR **326** are at the low state.

[0065] A transition of the ro_enable signal to the high state, causes the output of the fourth XOR **340** on the signal path **342** to transition to the high state. This, in turn, causes the output of the first XOR **326** on the signal path **328** to transition to the high state. The resulting rising edge at the CLK input of the first FF **302** causes the first FF **302** to latch the value at its D input (high state) to its Q output. Thus, the signal path **308** transitions to the high state. This causes the output of the first inverter **312** (D input of the first FF **302**) to transition to the low state.

[0066] The transition of the signal path **308** to the high state causes the output of the second XOR

330 on the signal path **332** to transition to the high state. The resulting rising edge at the CLK input of the second FF **304** causes the second FF **304** to latch the value at its D input (high state) to its Q output. Thus, the signal path **314** transitions to the high state. This causes the output of the second inverter **318** (D input of the second FF **304**) to transition to the low state.

[0067] The transition of the signal path **314** to the high state causes the output of the third XOR **334** on the signal path **336** to transition to the high state. The resulting rising edge at the CLK input of the third FF **306** causes the third FF **306** to latch the value at its D input (high state) to its Q output. Thus, the signal path **320** transitions to the high state. This causes the output of the third inverter **324** (D input of the third FF **306**) to transition to the low state.

[0068] The signal path **320** is fed back to the fourth XOR **340**. The transition of the signal path **320** to the high state causes the output of the fourth XOR **340** on the signal path **342** to transition to the low state (both inputs to the fourth XOR **340** are now at the high state). The transition of the signal path **342** to the low state causes the output of the first XOR **326** on the signal path **328** to transition to the high state since the signal path **308** is now at the high state.

[0069] The resulting rising edge at the CLK input of the first FF **302** causes the first FF **302** to latch the value at its D input (low state) to its Q output. Thus, the signal path **308** transitions to back to the low state. This causes the output of the first inverter **312** (D input of the first FF **302**) to transition back to the high state.

[0070] Thus, it may be seen that the outputs of the FFs will repeatedly transition between the low state and the high state. Consequently, the resulting oscillating signal of the RO **300** can be output via a buffer **344** as an ro_out signal.

[0071] The delay characteristics of ROs such as the RO **300** across an IC die may vary according to process variations across the IC die. An example of such a delay characteristic is the clock-to-Q time of an edge-triggered memory device. This delay time may be referred to herein as the clock-to-Q arc. Thus, the clock-to-Q arc may be shorter for some ROs than for other ROs on an IC die due to process variations.

[0072] The operating frequency of the RO **300** is based on the CLK-to-Q rise and fall arcs and the delays of the XORs. Since these parameters may be affected by process variations, the operating frequency of the RO **300** may be affected by process variations. Consequently, different ROs on an IC die may have different operating frequencies due to process variations. Thus, the RO **300** may be used as a sensor for power supply voltage scaling (e.g., as discussed above in conjunction with FIG. 2).

[0073] The FFs of the RO **300** do not need to include internal reset functionality to reset the FFs back to a given state for each oscillation cycle. Consequently, since the FF design of the RO **300** may be simpler than FFs that include internal reset functionality, the FF may take up less area on the IC die. This frees up additional IC die area that may be used, for example, to provide additional functionality that might not otherwise be provided on the IC die.

[0074] Also, in contrast with a FF design that incorporates internal reset functionality, the frequency of the RO **300** is based on clock-to-Q arcs for both low to high transitions of the Q outputs and high to low transitions of the Q outputs. Thus, the RO **300** may be used to obtain a more robust (e.g., more accurate) characterization of the FF delay characteristics based on both clock-to-Q arcs as compared to other RO designs. For example, an RO design that uses FFs with internal reset functionality may characterize the FF delay only with respect to one of the clock-to-Q arcs (e.g., for the low to high transitions of the Q outputs).

[0075] In some examples, an RO may include initialization circuitry to set all of the FFs of the RO to a known state. An example of such circuitry is shown in the example RO **400** of FIG. 4.

[0076] The RO **400** is depicted as including three stages similar to the RO **300**. Again, a different number of stages may be employed in different implementations.

[0077] The RO **400** includes the same basic design as the RO **300** of FIG. 3, except that the logic circuits of the RO **400** are represented as logic elements as would be implemented, for example, on

an IC die. A first FF **402**, a second FF **404**, and a third FF **406** of FIG. **4** correspond to the first FF **302**, the second FF **304**, and the third FF **306** of FIG. **3**, respectively. Similarly, a first XOR **426**, a second XOR **430**, a third XOR **434**, and a fourth XOR **440** of FIG. **4** correspond to the first XOR **326**, the second XOR **330**, the third XOR **334**, and the fourth XOR **340** of FIG. **3**, respectively. A first negative OR (NOR) **412**, a second NOR **418**, and a third NOR **424** provide an inversion function during operation similar to the functionality of the first inverter **312**, the second inverter **318**, and the third inverter **324** of FIG. **3**, respectively. A signal path **408**, a signal path **414**, a signal path **420**, a signal path **428**, a signal path **432**, a signal path **436**, and a signal path **442** correspond to the signal path **308**, the signal path **314**, the signal path **320**, the signal path **328**, the signal path **332**, the signal path **336**, and the signal path **342** of FIG. **3**, respectively. A buffer **444** corresponds to the buffer **344** of FIG. **3**.

[0078] Initialization circuitry including a pulse generator circuit **450**, a reset circuit **452**, a first multiplexer **454**, a second multiplexer **456**, and a third multiplexer **458** serve to reset the RO **400** to a known state prior to oscillation of the RO **400** being enabled. Specifically, when invoked, the initialization circuitry resets each of the inputs and the outputs of the first FF **402**, the second FF **404**, and the third FF **406** to a low state. This initialization circuitry is not in the frequency-determining RO oscillation signal path. Thus, the initialization circuitry does not impact the operating frequency of the RO **400**.

[0079] The pulse generator circuit **450** generates an initial pulse (initial_pulse) signal that is output by a buffer **460** on a signal path **462**. The initial_pulse signal is fed to a “b” input of each of the first multiplexer **454**, the second multiplexer **456**, and the third multiplexer **458**. The pulse generator circuit **450** may generate the initial_pulse signal in different ways in different examples. In some examples, the pulse generator circuit **450** includes circuitry to selectively enable a clock signal onto the signal path **462**. This clock signal may be generated by the pulse generator circuit **450** or some other circuit (e.g., a clock generator of an SoC). In some examples, the pulse generator circuit **450** may include circuitry to toggle a register, an I/O pin, or some other circuit to generate the initial_pulse signal. In some examples, at least some of the functionality of the pulse generator circuit **450** may be implemented in a processor (e.g., one of the processors of FIG. **1**).

[0080] The reset circuit **452** generates a reset signal that is output by a buffer **464** on a signal path **466**. The reset signal is fed to the control input of each of the first multiplexer **454**, the second multiplexer **456**, and the third multiplexer **458**, and to a respective input of each of the first NOR **412**, the second NOR **418**, and the third NOR **424**. The reset circuit **452** may generate the reset signal in different ways in different examples. In some examples, the reset circuit **452** may include circuitry to toggle a register, an I/O pin, or some other circuit to generate the reset signal. In some examples, at least some of the functionality of the reset circuit **452** may be implemented in a processor (e.g., one of the processors of FIG. **1**).

[0081] An enable circuit **468** generates an ro_enable signal that is output on a signal path **470** to an input of the fourth XOR **440**. The enable circuit **468** may generate the ro_enable signal in different ways in different examples. In some examples, the enable circuit **468** may include circuitry to toggle a register, and I/O pin, or some other circuit to generate the ro_enable signal. In some examples, at least some of the functionality of the enable circuit **468** may be implemented in a processor (e.g., one of the processors of FIG. **1**).

[0082] An example of an initialization operation follows. Other initialization operations may be used in other examples.

[0083] The enable circuit **468** sets the ro_enable signal to a low state whenever the oscillation of the RO **400** is being disabled. This is represented in FIG. **4** by the logic value “0” (e.g., a voltage level at or near ground) next to the signal path **470**.

[0084] To commence the initialization operation, the reset circuit **452** sets the reset signal to the high state. This is represented in FIG. **4** by the large logic value “1” (e.g., a voltage level at or near VDD) next to the signal path **466**.

[0085] When the reset signal is at the high state, the value at the data input of each of the FFs will be at the low state. Specifically, the signals d_0, d_1, and d_2 output by the first NOR **412**, the second NOR **418**, and the third NOR **424**, respectively, will each be at the low state as represented by the corresponding logic values of “0” in FIG. **4**.

[0086] In addition, the reset signal at the high state means that the multiplexer control inputs will be at the high state. Thus, each of the multiplexers will select its “b” input as its output. Thus, the signal fb_0 output by the first multiplexer **454** on a signal path **472** will correspond to the initial_pulse signal. The signal fb_1 output by the second multiplexer **456** on a signal path **474** will correspond to the initial_pulse signal. The signal fb_2 output by the third multiplexer **458** on a signal path **476** will correspond to the initial_pulse signal.

[0087] In conjunction with the reset signal being at the high state, the pulse generator circuit **450** activates the initial_pulse signal. In some examples the initial_pulse signal consists of several pulses. The initial_pulse signal is represented in FIG. **4** by the short rectangular wave next to the signal path **462**.

[0088] Each of the first XOR **426**, the second XOR **430**, and the third XOR **434** passes the initial_pulse signal from its respective multiplexer (signals fb_0, fb_1, and fb_2) to its respective output. Thus, each of the clock signals clk_0, clk_1, and clk_2 will include a rising edge that causes each of the first FF **402**, the second FF **404**, and the third FF **406** to propagate its D input to its Q output. Consequently, each of the output signals zn_1, zn_2, and zn_3 will be at the low state as indicated.

[0089] At the end of the initialization operation, the reset circuit **452** sets the reset signal to the low state. At this point, the RO **400** is initialized to a desired state (all FF outputs at the low state and all FF inputs at the high state).

[0090] FIG. **5** illustrates an example of an initial state of the RO **400** of FIG. **4** at the end of the initialization operation. As mentioned above, since the reset signal is now at the low state, the first NOR **412**, the second NOR **418**, and the third NOR **424** function as inverters. Thus, the signals d_0, d_1, and d_2 are set to the high state.

[0091] In addition, the reset signal at the low state means that the multiplexer control inputs will be at the low state. Thus, each of the multiplexers will select its “a” input as its output. Consequently, the signal fb_0 output by the first multiplexer **454** on a signal path **472** will correspond to the output signal zn_1. The signal fb_1 output by the second multiplexer **456** on a signal path **474** will correspond to the output signal zn_2. The signal fb_2 output by the third multiplexer **458** on a signal path **476** will correspond to the output signal zn_3.

[0092] As shown in FIG. **5**, all of the inputs to the XORs are at the low state. Thus, each of the clock signals clk_0, clk_1, and clk_2 is at the low state.

[0093] With the above in mind, several operational states of the RO **400** will now be described with reference to FIGS. **6-13**. These figures illustrate, in sequence, state transitions that follow the initialization operation.

[0094] FIG. **6** illustrates an example of a first operational state of the RO **400** of FIG. **4** that involves enabling the RO **400**. Initially, the enable circuit **468** sets the enable signal to a high state. This results in a rising edge propagating to the clock signal clk_0.

[0095] FIG. **7** illustrates an example of a second operational state of the RO **400** of FIG. **4** following the first operational state of FIG. **6**. The rising edge the clock signal clk_0 (as shown in FIG. **6**) caused the first FF **402** to capture the value at its D input (d_0=high state in FIG. **6**) and propagate that value to its Q output. Thus, the output signal zn_1 transitions to a high state. This results in a rising edge propagating to the clock signal clk_1.

[0096] When the output signal zn_1 transitions to the high state, the signal fb_0 transitions to the high state as well. This causes the clock signal clk_0 to transition to the low state. In addition, the input signal d_0 transitions to the low state.

[0097] Thus, by feeding back the output signal zn_1 to the D and CLK inputs of the first FF **402**,

both of these inputs are reset to the low state. Consequently, the first FF **402** will be ready to receive another clock pulse as discussed below that will cause the first FF **402** to toggle its output again.

[0098] FIG. **8** illustrates an example of a third operational state of the RO **400** of FIG. **4** following the second operational state of FIG. **7**. The rising edge the clock signal clk_1 (as shown in FIG. **7**) caused the second FF **404** to capture the value at its D input (d_1=high state in FIG. **7**) and propagate that value to its Q output. Thus, the output signal zn_2 transitions to a high state. This results in a rising edge propagating to the clock signal clk_2.

[0099] When the output signal zn_2 transitions to the high state, the signal fb_1 transitions to the high state as well. This causes the clock signal clk_1 to transition to the low state. In addition, the input signal d_1 transitions to the low state.

[0100] Thus, by feeding back the output signal zn_2 to the D and CLK inputs of the second FF **404**, both of these inputs are reset to the low state. Consequently, the second FF **404** will be ready to receive another clock pulse.

[0101] FIG. **9** illustrates an example of a fourth operational state of the RO **400** of FIG. **4** following the third operational state of FIG. **8**. The rising edge the clock signal clk_2 (as shown in FIG. **8**) caused the third FF **406** to capture the value at its D input (d_2=high state in FIG. **8**) and propagate that value to its Q output. Thus, the output signal zn_3 transitions to a high state.

[0102] When the output signal zn_3 transitions to the high state, the signal fb_2 transitions to the high state as well. This causes the clock signal clk_2 to transition to the low state. In addition, the input signal d_2 transitions to the low state.

[0103] Thus, by feeding back the output signal zn_3 to the D and CLK inputs of the third FF **406**, both of these inputs are reset to the low state. Consequently, the third FF **406** will be ready to receive another clock pulse.

[0104] As discussed herein, the output of the final stage of the RO **400** is fed back to the first stage to enable oscillation of the RO **400**. As shown in FIG. **9**, when the output signal zn_3 transitions to the high state, the signal zn_0 transitions to the low state. This results in a rising edge propagating to the clock signal clk_0.

[0105] FIG. **10** illustrates an example of a fifth operational state of the RO **400** of FIG. **4** following the fourth operational state of FIG. **9**. The rising edge the clock signal clk_0 (as shown in FIG. **9**) caused the first FF **402** to capture the value at its D input (d_0=low state in FIG. **9**) and propagate that value to its Q output. Thus, the output signal zn_1 transitions to a low state. This results in a rising edge at the clock signal clk_1.

[0106] When the output signal zn_1 transitions to the low state, the signal fb_0 transitions to the low state as well. This causes the clock signal clk_0 to transition to the low state. In addition, the input signal d_0 transitions to the high state.

[0107] By feeding back the output signal zn_1 to the CLK input of the first FF **402**, the clock signal clk_0 is reset to the low state. In addition, by feeding back the output signal zn_1 to the D input of the first FF **402**, the signal d_0 is set to the high state. Consequently, the first FF **402** will be ready to receive another clock pulse that will cause the first FF **402** to again toggle its output.

[0108] The low state will be propagated to the second FF **404** and the third FF **406** in a similar manner as discussed above. Then, when the low state propagates back to the fourth XOR **440**, the high state will again be propagated through the first FF **02**, the second FF **404**, and the third FF **406** as discussed above in conjunction with FIGS. **6-10**. As such, the RO **400** will continue to oscillate until the RO **400** is disabled.

[0109] FIG. **11** illustrates an example of disabling the RO **400** of FIG. **4**. The enable circuit **468** sets the ro_enable signal to a low state. The reset circuit sets the reset signal to a high state. The pulse generator circuit **450** sets the initial_pulse signal to a low state. Thus, all of the D inputs of the FFs are set to the low state and pulses cannot propagate through the stages. Consequently, the RO **400** ceases oscillation.

[0110] The RO **400** may be implemented in different ways in different examples. For example, in an alternative implementation, the reset function could set the input and the output of every FF to a high state. In this case, each of the NORs could be replaced with a corresponding negative AND (NAND) to provide the Q-to-D inversion function. The reset circuit **452** could drive the reset signal on the signal path **466** to a low state to perform the reset function in this case (to force the FFs inputs to the high state). Also, the inputs to the multiplexers may be switched so that each multiplexer outputs the initial_pulse signal on the signal path **462** when the reset signal is at the low state. Alternatively, inverters may be employed at the NAND reset inputs whereby the reset circuit **452** would control the reset signal as discussed in FIG. **4** (high state corresponds to reset being activated).

[0111] FIG. **12** illustrates an example timing diagram **1200** for the RO **400** of FIG. **4**. Initially, the initial_pulse signal is at a low state, the reset signal is at a high state, and the ro_enable signal is at a low state. Subsequently, the initial_pulse signal is set to the high state, then to the low state which causes the RO **400** to be set to an initialized state. For example, as shown in FIG. **12**, the ro_out signal as well as the clock signals clk_0, clk_1, and clk_2 are set to the low state. The reset signal is then set to the low state and the ro_enable signal is set to the high state, enabling oscillation of the RO **400**. That is, the ro_out signal and the clock signals clk_0, clk_1, and clk_2 repeatedly transition between the low state and the high state based on the inherent oscillation frequency of the RO **400** (e.g., based the FF clock-to-Q arcs and the XOR delays). The RO **400** continues to oscillate until the ro_enable signal is set to the low state.

[0112] FIG. **13** illustrates a multi-stage RO **1300** in accordance with certain aspects of the disclosure. The multi-stage RO **1300** includes a first stage **1302**, a final stage **1304**, and any number of intervening stages **1306**. The intervening stages **1306** are coupled to the first stage **1302** and the final stage **1304** as represented by the ellipses **1308**.

[0113] As used herein, the term “coupled” or “coupling” can refer to a direct coupling or an indirect coupling. An example of an indirect coupling is “coupled via one or more circuits.” For example, the Q output **1310** of the first stage **1302** is coupled to the CLK input **1312** of the final stage **1304**. In this case, the coupling is indirect in that the coupling is via one or more circuits of each stage of the intervening stages **1306**.

[0114] Each stage of the multi-stage RO **1300** includes an edge-triggered storage device, an internal feedback path for data (D), and an internal feedback path for clock (CLK). For example, the first stage **1302** includes a first edge-triggered memory device **1314** (e.g., a FF, a register, etc.), a first feedback path for data **1316** including an inversion circuit (not shown), and a first feedback path for clock **1318** (e.g., including a multiplexer and other logic, not shown). Similarly, the final stage **1304** includes a second edge-triggered memory device **1320** (e.g., a FF, a register, etc.), a second feedback path for data **1322** including an inversion circuit (not shown), and a second feedback path for clock **1324** (e.g., including a multiplexer and other logic, not shown). Also, each stage of the intervening stages **1306** includes a third edge-triggered memory device **1326** (e.g., a FF, a register, etc.), a third feedback path for data **1328** including an inversion circuit (not shown), and a third feedback path for clock **1330** (e.g., including a multiplexer and other logic, not shown).

[0115] The multi-stage RO **1300** also includes a feedback path for clock **1332** that couples the Q output of the final stage **1304** to the CLK input of the first stage **1302**. As discussed herein, this coupling may be an indirect coupling in that the coupling may be via one or more logic circuits (e.g., XOR gates).

[0116] FIG. **14** illustrates an example ring oscillator method **1400** in accordance with certain aspects of the disclosure. As described below, some or all illustrated features may be omitted in a particular implementation within the scope of the present disclosure, and some illustrated features may not be required for implementation of all examples. In some examples, the method **1400** may be carried out by the apparatus **200** illustrated in FIG. **2**, the RO **400** illustrated in FIG. **5**, the multi-stage RO **1300** illustrated in FIG. **13**, an SoC, or a processing system. In some examples, the

method **1400** may be carried out by any suitable apparatus or means for carrying out the functions or algorithm described below.

[0117] At block **1402**, an apparatus resets a ring oscillator. In some examples (e.g., as discussed above in conjunction with FIG. **4**), the apparatus may set the clock input and data output of each storage device at each stage of the ring oscillator to zero (e.g., a “0” logic state). In other examples (e.g., as discussed above for the alternative implementation), the apparatus may set the clock input and data output of each storage device at each stage of the ring oscillator to one (e.g., a “1” logic state). In some examples, the reset circuit **452** and/or the pulse generator circuit **450** of FIG. **4** may provide a means to reset a ring oscillator.

[0118] At block **1404**, the apparatus strobes a clock input of a first storage device at first stage of the ring oscillator. In some examples, the enable circuit **468** of FIG. **4** may provide a means to strobe a clock input of a first storage device at first stage of the ring oscillator.

[0119] At block **1406**, the apparatus feeds back the output of the first storage device to the clock input of the first storage device. In some examples, the first multiplexer **454** and the first XOR **426** of FIG. **4** may provide a means to feedback the output of the first storage device to clock input of the first storage device. In some examples, the first feedback path for clock **1318** of FIG. **13** may provide a means to feedback the output of the first storage device to clock input of the first storage device.

[0120] At block **1408**, the apparatus serially propagates the output of the first storage device to each storage device at each stage of the ring oscillator, while feeding-back the output of each storage device to the clock input of the corresponding storage device. In some examples, the circuitry associated with the second FF **404** and the third FF **406** of FIG. **4** may provide a means to serially propagate the output of the first storage device to each storage device at each stage of the ring oscillator, while feeding-back the output of each storage device to the clock input of the corresponding storage device. In some examples, the intervening stages **1306** and the final stage **1304** of FIG. **13** may provide a means to serially propagate the output of the first storage device to each storage device at each stage of the ring oscillator, while feeding-back the output of each storage device to the clock input of the corresponding storage device.

[0121] At block **1410**, the apparatus feeds back the output of the storage device at the final stage of the ring oscillator to the clock input of the first storage device. In some examples, the fourth XOR **440** and the first XOR **426** of FIG. **4** may provide a means to feedback the output of the storage device at the final stage of the ring oscillator to the clock input of the first storage device. In some examples, the feedback path for clock **1332** of FIG. **13** may provide a means to feedback the output of the storage device at the final stage of the ring oscillator to the clock input of the first storage device.

[0122] At block **1412**, the operations of blocks **1404** through **1410** are repeated until the ring oscillator is disabled. In some examples, the enable circuit **468** of FIG. **4** may provide a means to disable the ring oscillator.

[0123] The following provides an overview of various aspects of the present disclosure.

[0124] Aspect 1: An apparatus, comprising: a first edge-triggered storage device comprising a first clock input, a first data input, and a first data output; a first feedback path from the first data output to the first clock input; a second edge-triggered storage device comprising a second clock input, a second data input, and a second data output, the second clock input being coupled to the first data output; a second feedback path from the second data output to the second clock input; and a third feedback path from the second data output to the first clock input.

[0125] Aspect 2: The apparatus of aspect 1, wherein: first signaling via the first feedback path repeatedly sets the first clock input to a low state; and second signaling via the second feedback path repeatedly sets the second clock input to the low state.

[0126] Aspect 3: The apparatus of any of aspects 1 through 2, wherein third signaling via the third feedback path repeatedly sets the first clock input to a high state.

[0127] Aspect 4: The apparatus of any of aspects 1 through 3, wherein: the first edge-triggered storage device comprises a first flip flop circuit; and the second edge-triggered storage device comprises a second flip flop circuit.

[0128] Aspect 5: The apparatus of any of aspects 1 through 3, wherein: the first edge-triggered storage device comprises a first data register circuit; and the second edge-triggered storage device comprises a second data register circuit.

[0129] Aspect 6: The apparatus of any of aspects 1 through 5, wherein: the first edge-triggered storage device is configured as a first stage of a ring oscillator; and the second edge-triggered storage device is configured as a final stage of the ring oscillator.

[0130] Aspect 7: The apparatus of aspect 6, wherein an oscillation frequency of the ring oscillator is based at least on: first clock-to-output delays associated with first rising edges and first falling edges output from the first data output of the first edge-triggered storage device; and second clock-to-output delays associated with second rising edges and second falling edges output from the second data output of the second edge-triggered storage device.

[0131] Aspect 8: The apparatus of aspect 7, further comprising: a voltage controller configured to scale a power supply voltage based at least in part on the oscillation frequency of the ring oscillator.

[0132] Aspect 9: The apparatus of any of aspects 1 through 8, wherein: the first edge-triggered storage device is configured as a first stage of a ring oscillator; and the second edge-triggered storage device is configured as a second stage of the ring oscillator.

[0133] Aspect 10: The apparatus of any of aspects 1 through 9, further comprising: at least a third edge-triggered storage device coupling the first data output to the second clock input.

[0134] Aspect 11: The apparatus of any of aspects 1 through 10, further comprising: a reset circuit that sets the first data output and the second data output to a low state.

[0135] Aspect 12: An apparatus, comprising: first storage means comprising a first clock input and a first data output; first feedback means that feedbacks a first signal from the first data output to the first clock input; second storage means comprising a second clock input and a second data output, the second clock input being coupled to the first data output; second feedback means that feedbacks a second signal from the second data output to the second clock input; and third feedback means that feedbacks a third signal from the second data output to the first clock input.

[0136] Aspect 13: A ring oscillator, comprising: a first edge-triggered storage device comprising a first clock input, a first data input, and a first data output; a first logic circuit comprising a first input, a second input, and a first output, the first input being coupled to the first data output of the first edge-triggered storage device, the second input being coupled to a reset circuit, the first output being coupled to the first data input of the first edge-triggered storage device; a first multiplexer comprising a first multiplexer input, a second multiplexer input, a control input, and a multiplexer output, the first multiplexer input being coupled to the first data output of the first edge-triggered storage device, the control input being coupled to the reset circuit; and a second logic circuit comprising a third input, a fourth input, and a second output, the third input being coupled to the multiplexer output, the second output being coupled to the first clock input of the first edge-triggered storage device.

[0137] Aspect 14: The apparatus of aspect 13, wherein: the first logic circuit comprises a negative OR (NOR) circuit or a negative AND (NAND) circuit; and the second logic circuit comprises an exclusive OR (XOR) circuit.

[0138] Aspect 15: The apparatus of any of aspects 13 through 14, wherein the first edge-triggered storage device comprises a flip flop circuit or a data register circuit.

[0139] Aspect 16: The apparatus of any of aspects 13 through 15, wherein the second multiplexer input is coupled to a pulse generator circuit.

[0140] Aspect 17: The apparatus of any of aspects 13 through 16, further comprising: a third logic circuit comprising a fifth input, a sixth input, and a third output, the fifth input being coupled to an enable circuit, the third output being coupled to the fourth input of the second logic circuit.

[0141] Aspect 18: The apparatus of aspect 17, wherein the third logic circuit comprises an exclusive OR (XOR) circuit.

[0142] Aspect 19: The apparatus of any of aspects 17 through 18, further comprising: a second edge-triggered storage device comprising a second clock input, and a second data output, the second data output being coupled to the sixth input of the third logic circuit; and a fourth logic circuit comprising a seventh input, an eighth input, and a fourth output, the seventh input being coupled to the first data output of the first edge-triggered storage device, the eighth input being coupled to the second data output of the second edge-triggered storage device, the fourth output being coupled to the second clock input of the second edge-triggered storage device.

[0143] Aspect 20: The apparatus of any of aspects 1 through 19, further comprising: at least a third edge-triggered storage device coupling the first data output of the first edge-triggered storage device to the seventh input of the fourth logic circuit.

[0144] Within the present disclosure, the term “aspects” does not require that all aspects of the disclosure include the discussed feature, advantage or mode of operation. The term “coupled” is used herein to refer to the direct or indirect coupling between two objects. For example, if object A physically touches object B, and object B touches object C, then objects A and C may still be considered coupled to one another-even if they do not directly physically touch each other. For instance, a first object may be coupled to a second object even though the first object is never directly physically in contact with the second object. The terms “circuit” and “circuitry” are used broadly, and intended to include both hardware implementations of electrical devices and conductors that, when connected and configured, enable the performance of the functions described in the present disclosure, without limitation as to the type of electronic circuits, as well as software implementations of information and instructions that, when executed by a processor, enable the performance of the functions described in the present disclosure. As used herein, the term “determining” may include, for example, ascertaining, resolving, selecting, choosing, establishing, calculating, computing, processing, deriving, investigating, looking up (e.g., looking up in a table, a database or another data structure), and the like. Also, “determining” may include receiving (e.g., receiving information), accessing (e.g., accessing data in a memory), and the like.

[0145] The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but are to be accorded the full scope consistent with the language of the claims, wherein reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Unless specifically stated otherwise, the term “some” refers to one or more. A phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover: a; b; c; a and b; a and c; b and c; and a, b, and c. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims.

Claims

1. An apparatus, comprising: a first edge-triggered storage device comprising a first clock input, a first data input, and a first data output; a first feedback path from the first data output to the first clock input; a second edge-triggered storage device comprising a second clock input, a second data input, and a second data output, the second clock input being coupled to the first data output; a second feedback path from the second data output to the second clock input; and a third feedback

path from the second data output to the first clock input.

2. The apparatus of claim 1, wherein: first signaling via the first feedback path repeatedly sets the first clock input to a low state; and second signaling via the second feedback path repeatedly sets the second clock input to the low state.

3. The apparatus of claim 2, wherein third signaling via the third feedback path repeatedly sets the first clock input to a high state.

4. The apparatus of claim 1, wherein: the first edge-triggered storage device comprises a first flip flop circuit; and the second edge-triggered storage device comprises a second flip flop circuit.

5. The apparatus of claim 1, wherein: the first edge-triggered storage device comprises a first data register circuit; and the second edge-triggered storage device comprises a second data register circuit.

6. The apparatus of claim 1, wherein: the first edge-triggered storage device is configured as a first stage of a ring oscillator; and the second edge-triggered storage device is configured as a final stage of the ring oscillator.

7. The apparatus of claim 6, wherein an oscillation frequency of the ring oscillator is based at least on: first clock-to-output delays associated with first rising edges and first falling edges output from the first data output of the first edge-triggered storage device; and second clock-to-output delays associated with second rising edges and second falling edges output from the second data output of the second edge-triggered storage device.

8. The apparatus of claim 7, further comprising: a voltage controller configured to scale a power supply voltage based at least in part on the oscillation frequency of the ring oscillator.

9. The apparatus of claim 1, wherein: the first edge-triggered storage device is configured as a first stage of a ring oscillator; and the second edge-triggered storage device is configured as a second stage of the ring oscillator.

10. The apparatus of claim 1, further comprising: at least a third edge-triggered storage device coupling the first data output to the second clock input.

11. The apparatus of claim 1, further comprising: a reset circuit that sets the first data output and the second data output to a low state.

12. An apparatus, comprising: first storage means comprising a first clock input and a first data output; first feedback means that feedbacks a first signal from the first data output to the first clock input; second storage means comprising a second clock input and a second data output, the second clock input being coupled to the first data output; second feedback means that feedbacks a second signal from the second data output to the second clock input; and third feedback means that feedbacks a third signal from the second data output to the first clock input.

13. A ring oscillator, comprising: a first edge-triggered storage device comprising a first clock input, a first data input, and a first data output; a first logic circuit comprising a first input, a second input, and a first output, the first input being coupled to the first data output of the first edge-triggered storage device, the second input being coupled to a reset circuit, the first output being coupled to the first data input of the first edge-triggered storage device; a first multiplexer comprising a first multiplexer input, a second multiplexer input, a control input, and a multiplexer output, the first multiplexer input being coupled to the first data output of the first edge-triggered storage device, the control input being coupled to the reset circuit; and a second logic circuit comprising a third input, a fourth input, and a second output, the third input being coupled to the multiplexer output, the second output being coupled to the first clock input of the first edge-triggered storage device.

14. The ring oscillator of claim 13, wherein: the first logic circuit comprises a negative OR (NOR) circuit or a negative AND (NAND) circuit; and the second logic circuit comprises an exclusive OR (XOR) circuit.

15. The ring oscillator of claim 13, wherein the first edge-triggered storage device comprises a flip flop circuit or a data register circuit.

- 16.** The ring oscillator of claim 13, wherein the second multiplexer input is coupled to a pulse generator circuit.
- 17.** The ring oscillator of claim 13, further comprising: a third logic circuit comprising a fifth input, a sixth input, and a third output, the fifth input being coupled to an enable circuit, the third output being coupled to the fourth input of the second logic circuit.
- 18.** The ring oscillator of claim 17, wherein the third logic circuit comprises an exclusive OR (XOR) circuit.
- 19.** The ring oscillator of claim 17, further comprising: a second edge-triggered storage device comprising a second clock input, and a second data output, the second data output being coupled to the sixth input of the third logic circuit; and a fourth logic circuit comprising a seventh input, an eighth input, and a fourth output, the seventh input being coupled to the first data output of the first edge-triggered storage device, the eighth input being coupled to the second data output of the second edge-triggered storage device, the fourth output being coupled to the second clock input of the second edge-triggered storage device.
- 20.** The ring oscillator of claim 19, further comprising: at least a third edge-triggered storage device coupling the first data output of the first edge-triggered storage device to the seventh input of the fourth logic circuit.
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